

GEO — Hidden Geometry Framework:
An Exploratory Geometric Interpretation
of Cosmological Suppression and Transfer Structures

Leonel Hernan Torreblanca
Independent Researcher
Argentina

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Abstract

This work presents GEO (Hidden Geometry Framework), an exploratory geometric interpretation of cosmological suppression, transfer structure behavior, and observational growth effects.

The framework investigates whether several observed cosmological quantities may emerge from an underlying geometric architecture not directly visible in standard formulations, but inferable through effective observational behavior.

A sequence of reproducible exploratory tests was developed using observationally motivated transfer fractions, suppression structures, and geometric consistency relations. The analysis explores preferred geometric regions associated with quantities such as $3/4$, $\sqrt{3/5}$, and $\pi/4$, alongside phenomenological transfer relations and effective growth suppression laws.

The presented results do not claim a replacement of standard cosmology, General Relativity, or Boltzmann evolution frameworks. Instead, GEO is proposed as an exploratory effective geometric layer intended for open validation, criticism, falsification, and independent reproduction.

All scripts, datasets, figures, and generated outputs are publicly available in the associated repository.

1. Introduction

Modern cosmology successfully describes large-scale observational phenomena through the Lambda Cold Dark Matter (LCDM) framework and Einstein-Boltzmann evolution systems.

However, several observational tensions and phenomenological structures motivate exploratory investigations into whether additional effective geometric interpretations may exist beneath the observable transfer behavior of cosmological structures.

The GEO framework emerged from a sequence of exploratory numerical experiments investigating whether suppression structures, growth fractions, and transfer efficiencies exhibit preferred geometric organization patterns.

The purpose of GEO is not to reject standard cosmology, but to investigate whether an additional effective geometric interpretation layer may describe part of the observed phenomenology.

This work presents the first public exploratory preprint of the GEO framework, including reproducible scripts, observationally motivated tests, generated figures, and phenomenological consistency relations.

2. Historical Development of GEO

The GEO framework emerged progressively through several exploratory stages.

Initial investigations began with generalized geometric decomposition ideas associated with hidden transfer structures and complementary suppression regions.

Early exploratory models evolved into the GDD framework, followed by later SOP exploratory architectures and finally the current GEO interpretation.

Throughout this process, the central recurring observation was the repeated emergence of stable geometric transfer fractions associated with observationally motivated suppression structures.

Several geometric quantities repeatedly appeared during the experiments, particularly:

- $3/4$
- $\sqrt{3/5}$
- $\pi/4$

These values emerged not as imposed constants, but as preferred effective geometric regions associated with transfer behavior, suppression structure, and architectural consistency relations.

The current GEO framework represents the first organized public presentation of these exploratory observations.

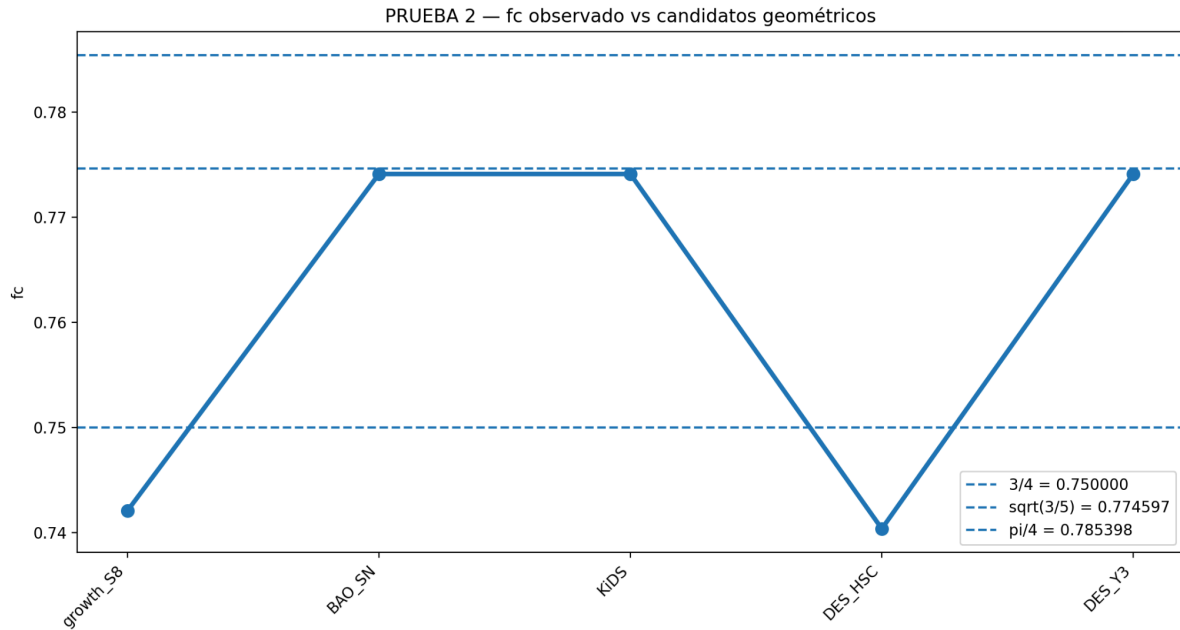


Figure 1 — Observed transfer fractions compared against preferred geometric candidate regions. The emergence of stable proximity to $\sqrt{3/5}$ appears repeatedly across multiple observationally motivated cases.

3. Geometric Interpretation

The central hypothesis explored by GEO is that part of the observed cosmological transfer behavior may correspond to an underlying effective geometric organization.

In this interpretation, the observable transfer fraction f_c behaves as an effective active geometric region, while the complementary fraction f_{out} represents an effective non-transfer or complementary geometric structure.

The exploratory tests repeatedly suggest that several observationally motivated cases stabilize near preferred geometric regions rather than exhibiting arbitrary distribution.

In particular, repeated proximity appears near:

- $3/4$
- $\sqrt{3/5}$
- $\pi/4$

The quantity $\sqrt{3/5}$ emerged repeatedly as an effective transitional geometric region associated with transfer stability and suppression behavior.

The framework does not currently interpret these quantities as fundamental constants, but rather as emergent effective geometric structures.

4. Effective Geometric Laws

Several phenomenological consistency relations emerged repeatedly during the exploratory tests.

One of the strongest recurrent relations was an effective geometric transfer law connecting suppression structures and transfer efficiency.

The most stable observed relation was:

$$R = \mu^{1/3}$$

where R represents an effective geometric transfer response and μ represents an effective suppression structure parameter.

The relation repeatedly produced stable proximity to observationally inferred transfer regions and generated low relative error across multiple exploratory configurations.

A second recurrent relation appeared in the effective geometric efficiency law:

$$\eta = f_c^2$$

where η represents an effective geometric efficiency associated with the active transfer region f_c .

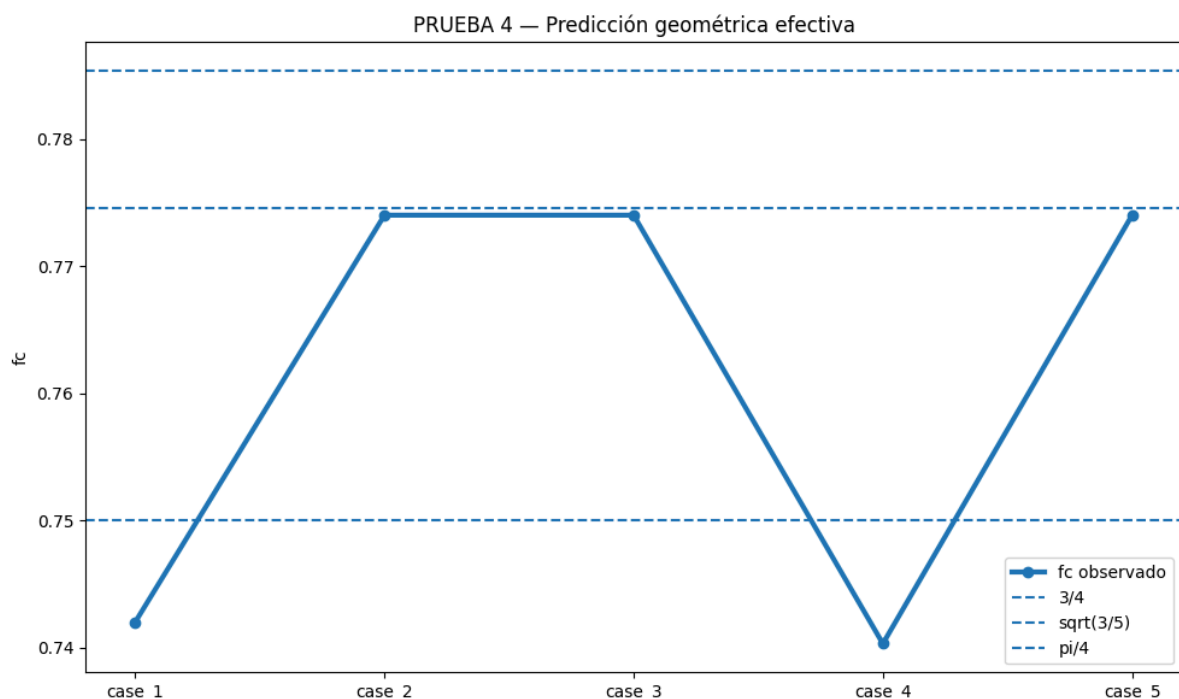


Figure 2 — Effective geometric prediction regions generated from the exploratory transfer law. Multiple observationally motivated cases converge near the $\sqrt{3/5}$ transitional region.

5. Exploratory Results

Across the exploratory GEO tests, several recurring behaviors appeared repeatedly.

The most stable observations included:

- emergence of preferred geometric transfer regions,
- effective suppression consistency near approximately 7% → 8%,
- stable transfer fractions near $\sqrt{3/5}$,
- and reproducible predictive transfer behavior.

The observed structures repeatedly emerged without requiring large arbitrary parameter explosions.

In particular, the stability of the effective geometric response relation:

$$R = \mu^{(1/3)}$$

suggested the possibility that observable suppression may possess an underlying geometric organization structure.

The framework currently interprets these results phenomenologically and geometrically rather than as established physical laws.

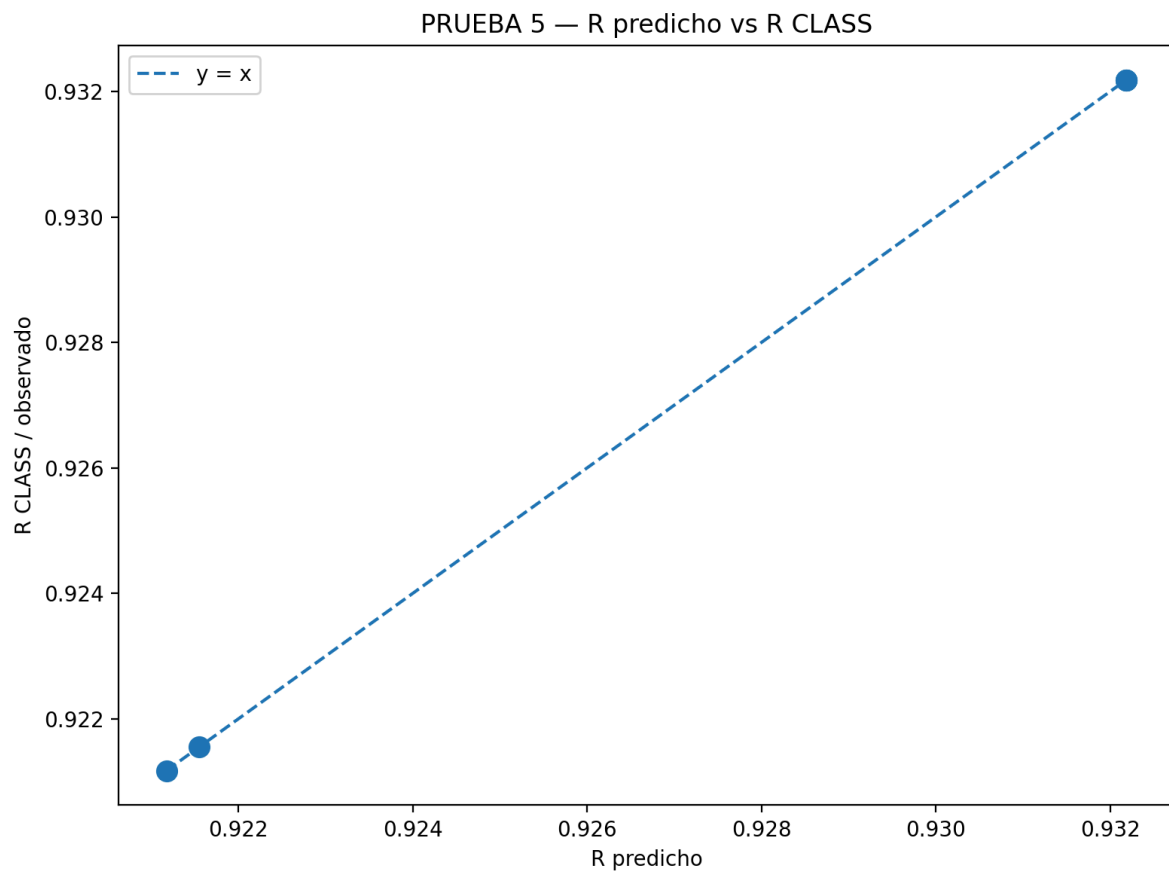


Figure 3 — Comparison between observed effective response and geometrically predicted response generated by the exploratory GEO law. The relation $R = \mu^{(1/3)}$ repeatedly produced low relative error across multiple observationally motivated configurations.

6. Limitations and Scientific Status

The GEO framework is currently exploratory.

The presented results do not constitute a validated replacement for Λ CDM, General Relativity, or Einstein-Boltzmann cosmological evolution.

The current analysis should instead be interpreted as:

- an exploratory geometric investigation,
- a phenomenological transfer framework,
- and a reproducible numerical research project.

The framework has not yet undergone peer review.

Several aspects remain incomplete and require future investigation, including:

- full Monte Carlo analysis,
- confrontation with complete cosmological likelihood pipelines,
- independent reproduction,
- systematic uncertainty analysis,
- and fully documented CLASS source-code modifications.

The purpose of the present public release is:

- transparency,
- open discussion,
- reproducibility,
- criticism,
- and independent falsification.

7. Falsifiability

The GEO framework is intended to remain fully open to independent verification and falsification.

Possible future falsification scenarios include:

- disappearance of the observed geometric stability under larger datasets,
- incompatibility with future cosmological likelihood analyses,
- instability under independent reproduction,
- or emergence of equivalent behavior through purely statistical artifacts.

The current repository intentionally provides:

- scripts,
- datasets,
- generated outputs,

- and geometric relations

to facilitate independent inspection and reproduction.

8. Conclusion

GEO proposes an exploratory hidden geometric interpretation of cosmological suppression and transfer behavior.

The framework emerged progressively through multiple exploratory stages involving geometric decomposition, suppression analysis, transfer structures, and effective predictive relations.

The repeated emergence of stable geometric regions near:

- $3/4$,
- $\sqrt{3/5}$,
- and $\pi/4$

alongside stable phenomenological transfer behavior motivated the development of the present framework.

The current public release provides:

- reproducible scripts,
- generated figures,
- datasets,
- documentation,
- and exploratory geometric analyses.

Future work should include:

- independent verification,
- Monte Carlo analysis,
- complete CLASS patch documentation,
- and confrontation against full cosmological likelihood pipelines.

The framework is presented openly as exploratory scientific research intended for validation, criticism, reproduction, or falsification by the broader scientific community.

Repository

Public repository:

<https://github.com/LeoTorreblanca/GEO-hidden-geometry-framework>

Author:

Leonel Hernan Torreblanca

Independent Researcher
Argentina